

Exercise 1 Stability and Invariant set

1. The closed-loop system is $x(k+1) = w(k)$ subject to the constraints $-x(k) \in \mathcal{U} = \{x \mid x \in [-10, 10]\} = \mathcal{X}$

The robust pre-set of $[-10, 10]$ is

$$\text{pre}^{\mathcal{W}}([-10, 10]) = \{x \mid x - x + w \in [-10, 10] \forall w \in \mathcal{W}\} = (-\infty, \infty)$$

$\text{pre}^{\mathcal{W}}([-10, 10]) \cap \mathcal{X} = (-\infty, \infty) \cap [-10, 10] = [-10, 10]$. Thus, the maximum robust invariant set $\mathcal{X}_f = [-10, 10]$.

2. Terminal set is

$$\bar{\mathcal{X}}_f = \mathcal{X}_f \ominus \left(\bigoplus_{i=0}^1 A^i \mathcal{W} \right) = \mathcal{X}_f \ominus 2\mathcal{W} = [-10, 10] \ominus [-2, 2] = [-8, 8]$$

3. Setting up QP formulation:

$$\begin{aligned} \min \quad & (u_0 \quad x_1 \quad u_1 \quad x_2) \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 5 \end{bmatrix} \begin{pmatrix} u_0 \\ x_1 \\ u_1 \\ x_2 \end{pmatrix} \\ \text{s.t.} \quad & \begin{bmatrix} 1 & -1 & 0 & 0 \\ 0 & 1 & 1 & -1 \end{bmatrix} \begin{pmatrix} u_0 \\ x_1 \\ u_1 \\ x_2 \end{pmatrix} = \begin{bmatrix} -1 \\ 0 \end{bmatrix} x_0 \\ & \begin{bmatrix} 1 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & -1 \end{bmatrix} \begin{pmatrix} u_0 \\ x_1 \\ u_1 \\ x_2 \end{pmatrix} \leq \begin{bmatrix} 10 \\ 10 \\ 10 \\ 10 \\ 8 \\ 8 \end{bmatrix} \end{aligned}$$

4. The problem will be infeasible for all states when the terminal set is empty. The terminal set is

$$\mathcal{X}_f \ominus \left(\bigoplus_{i=0}^{N-1} A^i \mathcal{W} \right) = \mathcal{X}_f \ominus N\mathcal{W} = (10 - N)[-1, 1].$$

This set is empty for all $N \geq 11$.

Exercise 2 **Pre-stabilizing control law**

1.

$$|u| \leq 1 \Leftrightarrow |Kx + v| \leq 1 \Leftrightarrow -1 \leq -1.5x + v \leq 1$$

2.

$$\begin{aligned} 0.5x_0 + v_0 - 0.5 &\leq x_1 \leq 0.5x_0 + v_0 + 0.5 \\ 0.25x_0 + 0.5v_0 + v_1 - 0.75 &\leq x_2 \leq 0.25x_0 + 0.5v_0 + v_1 + 0.75 \\ 0.125x_0 + 0.25v_0 + 0.5v_1 + v_2 - 0.875 &\leq x_3 \leq 0.125x_0 + 0.25v_0 + 0.5v_1 + v_2 + 0.875 \end{aligned}$$

3.

$$|x_3| \leq 3 \Leftrightarrow 0.125x_0 + 0.25v_0 + 0.5v_1 + v_2 \leq 3 - 0.875$$